

Quantitative Football Betting Pipeline

Technical Architecture, Feature Engineering & Risk Engine Specification

Prepared for: Christian | Project Group 11 | August 2026

1. END-TO-END PIPELINE WORKFLOW

Phase 1: LightGBM Classifier Training

The system trains a LightGBM gradient boosted classifier to predict multi-class outcome distributions across three target classes (0: Home Win, 1: Draw, 2: Away Win). The model is strictly restricted to pre-kickoff contextual and rolling performance features.

- **Strict Temporal Split:** To eliminate lookahead bias and data leakage, training is conducted on ~50,000 matches from 2015 to 2023. Model validation and out-of-sample backtesting are performed on ~15,000 matches spanning 2023 to 2026.

Phase 2: Probability Calibration (Isotonic Regression)

Raw machine learning probabilities are frequently miscalibrated at extreme bounds. We fit a 3-fold **Isotonic Regression** calibration layer over out-of-fold cross-validation predictions. This guarantees that when the calibrated model outputs a 60% probability, the outcome occurs in exactly 60% of cases over historical samples.

Phase 3: Pinnacle Benchmark & Log Loss Evaluation

We evaluate calibrated model probabilities against Pinnacle's fair de-vigged market probabilities (p_shin) on the 15,000 out-of-sample test matches using logarithmic loss:

- **Model Log Loss < Pinnacle Log Loss:** Model features extract unique alpha that beats bookmaker consensus.
- **Model Log Loss > Pinnacle Log Loss:** Pinnacle remains the sharpest global benchmark; system profit relies on selective edge discovery (betting only on mispriced discrepancies).

2. FEATURE ENGINEERING ARCHITECTURE

All input features are generated strictly prior to kickoff using rolling time-series calculations and exponential smoothing window functions.

Feature Category	Column Name	Technical Specification & Strategic Purpose
Rest Days & Fatigue	home_rest / away_rest	Days since previous fixture (capped at 14 days). Quantifies physical recovery and fatigue.
	diff_rest_days	home_rest - away_rest. Net rest advantage (e.g., +4 days indicates Home team is fresher).
Short-Term Form (5 Games)	home_GF_5 / away_GF_5	Exponentially weighted goals scored over past 5 games. Evaluates current attack momentum.
	home_GA_5 / away_GA_5	Exponentially weighted goals conceded over past 5 games. Evaluates defensive vulnerability.
	home_GD_5 / away_GD_5	GF_5 - GA_5. Net goal differential evaluating recent true performance quality.
Medium-Term Form (10 Games)	GF_10 / GA_10	Averaged over 10 games to smooth out short-term random noise and reveal true team baseline capability.
Differentials & Matchups	diff_roll_GF_5	home_GF_5 - away_GF_5. Positive values denote Home offensive advantage over Away attack.
	diff_roll_GA_5	home_GA_5 - away_GA_5. Negative values denote Home defensive superiority over Away defense.

3. COLUMN STANDARDIZATION & DATASET INTERPRETATION

Pinnacle & Bookmaker Mapping Standardization

Historical datasets across 11 seasons utilize varying naming conventions. The pipeline standardizes them into unified schema headers:

- PinnacleH / PH / PSH → **PSH** (Pinnacle Closing Home Odds)
- BbAvH → **AvGH** (BetBrain / Market Average Home Odds)
- BbMxH → **MaxH** (BetBrain / Market Maximum Home Odds)

Sample Data Row Interpretations

Match Details	Odds (PSH)	Shin Prob (p_shin_H)	Goal Diff (diff_roll_GF_5)	Pipeline Interpretation & Strategic Insight
Club Brugge vs. Kortrijk (2015-08-14)	1.43	0.6886 (68.86%)	+0.421	Heavy home favorite entering in superior attacking form (+0.42 goals/game advantage). Market and form align cleanly.
Genk vs. Westerlo (2015-08-15)	1.40	0.7005 (70.05%)	-1.316	Key Insight: Genk has inferior recent scoring (-1.316) yet market prices them as a massive 70% favorite. Demonstrates market's ability to price baseline quality over short-term streaks.
Mouscron vs. Standard Liège (2015-08-15)	5.10	0.1865 (18.65%)	-0.632	Mouscron is a longshot underdog (18.65% win chance). Standard Liège outscoring them significantly (-0.632 diff). Away dominant expected match.
Waasland-Beveren vs. Waregem (2015-08-15)	2.86	0.3399 (33.99%)	+0.105	Balanced contest. Home team slightly underdogged (33.99%) despite virtually identical scoring rates (+0.10). Tight fixture.
Lokeren vs. Mechelen (2015-08-15)	2.37	0.4118 (41.18%)	-0.789	Lokeren holds home-field price edge (41.18%), but Mechelen enters in stronger offensive form (+0.79 goals/game). Clash of home edge vs form.

4. EXPLAINING MODEL PERFORMANCE & BENCHMARK TO CHRISTIAN

Analogy for Christian: Master Meteorologist vs. Statistical Computer

Pinnacle is like a master weather forecaster with access to satellite imagery, local stations, barometers, and expert meteorologists. Their odds instantly reflect squad news, tactical shifts, weather, and sharp institutional capital right before kickoff.

LightGBM is like a smart automated algorithm relying purely on historical rainfall averages (5 to 10-game goal stats and rest days). It lacks real-time injury context, which is why its global log loss (1 . 0341) is slightly behind Pinnacle's benchmark (0 . 9937).

Strategic Core: Selective Edge Discovery

We do not need to beat Pinnacle on all 15,211 matches. We only need to spot localized pricing errors where the market over- or under-prices an outcome relative to pure statistical baselines. By placing bets strictly when our expected return exceeds strict thresholds, we ignore 95% of efficient matches and exploit the 5% where pricing discrepancies exist.

5. CORE INTUITIVE CONCEPTS EXPLAINED

Concept A: What is an Edge? (The Weighted Coin Analogy)

The Weighted Coin Analogy

Imagine a normal coin flip paying 1:1. Everyone assumes Heads lands 50% of the time. However, you privately test the coin and prove it lands on Heads **55%** of the time.

- **Market Consensus:** 50% chance
- **Your Knowledge:** 55% chance
- **Your Edge:** +5% mathematical advantage

You won't win every flip, but over 1,000 flips, that +5% edge guarantees long-term systematic profit while the market breaks even or loses to vigorish.

Application to Live Slate (Arsenal vs. Chelsea):

- Pinnacle true probability (p_{shin}): **59.27%**
- Model predicted probability (p_{model}): **62.92%**
- Calculated Edge: 62.92% - 59.27% = **+3.65% Edge**
- Offered Odds (\$1.95\$) are too high relative to actual probability, yielding a positive expected value trade.

Concept B: Soft vs. Sharp Bookmakers (The iPhone Analogy)

The iPhone Pricing Analogy

An iPhone costs \$1,000 everywhere online (Market Value). A local electronics shop has slow updates and tags the exact same phone for \$800. You immediately purchase the \$800 phone because you know its true value is \$1,000.

Bookmaker Type	Target Audience	Operational Behavior
Sharp Books (Pinnacle)	Professionals & Syndicates	Adjusts odds in seconds based on high-volume professional capital. Serves as our gold-standard true probability baseline.
Soft Books (Bet365, Retail)	Casual Bettors	Slow to react to market shifts. Creates pricing lags where odds remain at \$1.85 on outcomes actually worth \$1.67.

6. RISK ENGINE, BACKTESTING & POSITION SIZING

Strict Execution Rules

- 1. **Draw Exclusion:** Excludes Draw selections (D) to focus on Home/Away decisive victories, reducing outcome variance.
- 2. **Odds Ceiling (\$\le 3.50\$):** Rejects longshots above \$3.50\$ odds to shield the bankroll from high-variance drawdown tails.
- 3. **Edge & EV Qualification Gate:**
 - Probability Edge Gate: $p_{model} - p_{shin} \geq +0.020$ (+2.0% minimum edge).
 - Expected Value Gate: $(p_{model} * Odds) - 1.0 \geq +0.030$ (+3.0% minimum EV).

Position Sizing: Fractional Quarter-Kelly (\$0.25 imes\$)

To avoid the aggressive drawdowns of Full Kelly betting, position size is scaled using Quarter-Kelly with a strict 3.0 unit cap per match:

$$\text{Stake} = \text{Min}(0.25 * [\text{EV} / (\text{Odds} - 1)] * 100, 3.0 \text{ Units})$$

Backtest Execution Output Summary

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SECTION I: FRACTIONAL KELLY EV BACKTESTING
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Total Bets Placed      : 927
Total Units Staked     : 1616.43 units (Avg Stake: 1.74 units)
Win Rate               : 59.22%
Total Profit/Loss      : +56.24 units
Return on Investment    : +3.48%
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